Hydrolethalus Syndrome 1 (HYLS1): A Comprehensive Disease Characterization

Disease: Hydrolethalus Syndrome 1 (HLS1) · **MONDO:** MONDO:0009365 · **OMIM:** 236680 · **Causal gene: HYLS1** (OMIM 610693; HGNC-approved symbol HYLS1; NCBI GeneID 219844; chromosome 11q24.2)

Summary

Hydrolethalus Syndrome 1 is a rare, autosomal-recessive, perinatally lethal multiple-malformation syndrome caused by **biallelic loss-of-function–type variants in the HYLS1 gene** on chromosome 11q24.2. Despite its serial-number name, the disease belongs squarely to the family of **ciliopathies**: HYLS1 encodes a conserved centriole/basal-body protein, and its dysfunction disrupts the assembly and signaling of the primary cilium during early embryogenesis. The overwhelmingly most common disease allele worldwide is the Finnish founder missense variant **c.632A>G, p.Asp211Gly (D211G; NM_145014.2; rs104894232)**, first linked to the disease by Mee et al. in 2005 (PMID: 15843405).

The clinical phenotype is a devastating, recognizable constellation of midline and developmental malformations: hydrocephaly/hydranencephaly with absent midline brain structures, the pathognomonic **occipitoschisis ("keyhole") skull-base defect**, micrognathia, cleft lip/palate, polydactyly of the hands and feet, and defective lobation of the lungs. Affected fetuses are stillborn or die within hours to days of birth; there is no disease-modifying therapy, and management is limited to prenatal diagnosis, genetic counseling, and perinatal palliative care.

Mechanistically, work in *C. elegans*, *Drosophila*, mouse, and mammalian cells converges on a coherent causal chain: the D211G mutation **disrupts the HYLS1–CEP120 interaction**, displacing HYLS1 from the centriole distal end and destabilizing the centriolar microtubule wall; this **loss of centriole integrity impairs ciliary-gate (transition fibre/transition zone) assembly** and intraflagellar transport docking; the resulting **defect in primary-cilium biogenesis and signaling—most critically Hedgehog signaling, modulated via the ciliary lipid kinase PIPKIγ and GPR161/β-arrestin handling—produces the midline, neural, and**

limb malformations characteristic of the syndrome. This report documents each link in that chain with primary-literature evidence, and provides the epidemiological, genetic, diagnostic, prognostic, and model-organism detail needed to populate a disease knowledge-base entry.

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1. Disease Information

Hydrolethalus Syndrome 1 is an autosomal-recessive, perinatally lethal fetal malformation disorder. It was originally delineated in the Finnish population, where it is part of the "Finnish disease heritage." The condition is defined by a reproducible pattern of central nervous system (CNS) midline malformations, craniofacial anomalies, polydactyly, and visceral (notably pulmonary) defects.

Key identifiers:

Resource	Identifier
MONDO	MONDO:0009365
OMIM (phenotype)	236680 (Hydrolethalus syndrome 1)
OMIM (gene)	610693 (HYLS1)
Gene (NCBI)	GeneID 219844
Cytogenetic locus	11q24.2
Orphanet	Hydrolethalus syndrome (rare, lethal)
MeSH / ICD	Indexed under congenital malformation syndromes; ICD-10 Q87.8 (other specified congenital malformation syndromes) is typically applied clinically

Synonyms / alternative names: Hydrolethalus syndrome (HLS); HLS1; Salonen-Herva-Norio syndrome.

Information source type: The disease-level knowledge is derived from **aggregated disease-level resources** (OMIM, Orphanet, ClinVar, gnomAD) and from **individual/small-series clinical reports and fetal pathology studies**, rather than from large EHR cohorts (the disease is too rare and uniformly lethal for EHR-scale study).

2. Etiology

Primary cause — genetic. HLS1 is a monogenic Mendelian disorder caused by **biallelic pathogenic variants in HYLS1**. There is no infectious or environmental cause. The disease is fully genetically determined once two pathogenic alleles are inherited.

Genetic risk factors. The dominant risk factor is **carrier status for the Finnish founder allele D211G (c.632A>G)**. In gnomAD v4 exomes this variant has a global allele frequency of ~5.9×10^{-4} but is enriched ~27-fold in Finns (Finnish AF 8.57×10^{-3} vs non-Finnish European 3.19×10^{-4}; Finding F006). Non-Finnish pathogenic alleles exist — two French fetuses carried the Finnish D211G in trans with novel alleles **c.662G>C, p.Arg221Pro** and **c.613C>T, p.Arg205*** (PMID: 34212369). A large contiguous-gene deletion at 11q23.3–q25 encompassing HYLS1 is also catalogued as pathogenic (Finding F008).

Environmental risk factors / protective factors / gene–environment interactions. None established. As a fully penetrant recessive embryonic-lethal disorder, HLS1 has no recognized environmental, lifestyle, dietary, or infectious modifiers, and no protective alleles or behaviors are documented. The only "protective" measure is avoidance of two pathogenic alleles in a conceptus (i.e., reproductive genetics). *Not applicable / not available for this disease.*

3. Phenotypes

HLS1 presents as a prenatal/neonatal multiple-malformation syndrome. Severity is uniformly **severe (lethal)**; onset is **congenital/prenatal**; the course is non-progressive because death occurs perinatally.

Phenotype	Type	Frequency / note	Suggested HPO term
Hydrocephaly / hydranencephaly	Physical/ structural (CNS)	Core, near-universal	HP:0000238 Hydrocephalus; HP:0007238 Hydranencephaly
Absent midline brain structures	Structural (CNS)	Core	HP:0007370 Aplasia/ hypoplasia of the corpus callosum
Occipitoschisis ("keyhole" skull-base defect)	Structural (cranial)	**Pathognomonic**	HP:0002703 Abnormal skull base morphology (closest)
Immature cerebral cortex / abnormal radial glia	Histopathologic (CNS)	Described at fetal autopsy	HP:0002538 (cortical malformation)
Hypothalamic hamartoma	Structural (CNS)	Reported key finding	HP:0002902 Hypothalamic hamartoma
Micrognathia	Craniofacial	Core	HP:0000347 Micrognathia
Cleft lip / cleft palate (midline)	Craniofacial	Common	HP:0000175 Cleft palate; HP:0410030 Cleft lip
Polydactyly (hands and feet; pre-/postaxial)	Limb	Core	HP:0100259 Postaxial polydactyly; HP:0001177 Preaxial polydactyly of hand
Defective lung lobation	Visceral (respiratory)	Core	HP:0002101 Abnormal lung lobation
Abnormal genitalia	Genitourinary	In non-Finnish cases	HP:0000078 Abnormality of the genital system
Macrocephaly	Craniofacial	Reported	HP:0000256 Macrocephaly

Evidence. Honkala et al. describe "central nervous system (CNS) malformation such as hydrocephaly and absent midline structures of the brain, micrognathia, defective lobation of the lungs and polydactyly" and note "immature cerebral cortex, abnormalities in radial glial cells and hypothalamic hamartoma are among key findings in the CNS of HLS fetuses" (PMID: 19400947). Belengeanu et al. identify occipitoschisis as "the pathognomonic sign of the

syndrome" alongside "macrocephaly, midline cleft-lip, cleft palate, polydactyly of both hands and feet" (PMID: 22029171). Ghesh et al. expand the spectrum to "pre and postaxial hexadactyly and abnormal genitalia" in non-Finnish fetuses (PMID: 34212369). A recent case report describes "open-book cerebral hemispheres, defective lobulation of the lungs (one lobe on the left, two on the right)... and a specific keyhole-shaped defect in the skull base" (PMID: 39857086).

Quality of life. Not applicable in the conventional sense — the disease is lethal before or shortly after birth. The burden falls on families (pregnancy loss, bereavement) rather than on surviving patients.

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4. Genetic / Molecular Information

Causal gene. HYLS1 (HGNC symbol HYLS1; OMIM 610693; GeneID 219844), a 299-amino-acid protein of originally unknown function, located at 11q24.2. Mee et al. mapped HLS to a 1 cM interval on 11q23–25, narrowed by linkage-disequilibrium/haplotype analysis to a 476 kb region between PKNOX2 and DDX25, and identified the causal A>G transition in a novel transcript (FLJ32915), which they named HYLS1 (PMID: 15843405).

Pathogenic variants.

Variant (NM_145014.2)	Protein	Type	Classification	Note
c.632A>G (rs104894232)	p.Asp211Gly (D211G)	Missense	Pathogenic	Finnish founder allele
c.662G>C	p.Arg221Pro	Missense	Reported pathogenic	Non-Finnish (French)
c.613C>T	p.Arg205*	Nonsense	Reported pathogenic	Non-Finnish (French)
11q23.3–q25 deletion	—	Structural/ CNV	Pathogenic	Contiguous-gene deletion

ClinVar landscape (Finding F008). A sampled set of ~40 HYLS1 ClinVar records spans the full classification spectrum (~7 Pathogenic, ~13 Likely pathogenic, ~12 VUS, ~7 Likely benign, 1 conflicting). The founder D211G is classified **Pathogenic** (criteria provided, multiple submitters).

Allele frequency (Finding F006). gnomAD v4 exomes: D211G global AF 5.88×10^{-4} (859/1,461,894); Finnish AF 8.57×10^{-3} (458/53,420); non-Finnish European AF 3.19×10^{-4} — a ~26.9-fold Finnish enrichment.

Origin. Germline (biallelic, inherited). No somatic contribution.

Functional consequences. The D211G missense allele is a **functional loss-of-function with respect to ciliogenesis**. Dammermann et al. demonstrated that "a single amino acid change in human HYLS1 leads to a perinatal lethal disorder... and we show that this mutation impairs HYLS-1 function in ciliogenesis" (PMID: 19656802). Mee et al. found altered subcellular localization: wild-type polypeptide was cytoplasmic whereas mutant protein "became localized in distinct nuclear structures" (PMID: 15843405).

Modifier genes / epigenetics. No validated modifier genes or epigenetic mechanisms are established for HLS1. *Not available.*

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5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents are implicated in HLS1. The disorder is a pure Mendelian recessive condition. *Not applicable for this disease.*

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6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Biallelic HYLS1 variant (most often D211G)** → produces a HYLS1 protein that **fails to interact normally with CEP120** (demonstrated in mouse; PMID: 38496445 / PMID: 40009365).
2. Loss of the HYLS1–CEP120 interaction → **HYLS1 is displaced from the centriole distal end → failure to recruit/stabilize centriole inner-scaffold proteins → destabilization of the**

centriolar microtubule wall and degeneration of the centriole distal end (demonstrated, mouse).

3. Loss of centriole structural integrity → **impaired assembly of the basal body's distal appendages/transition fibres (TF) and transition zone (TZ) → defective ciliary gate** (demonstrated, *C. elegans*; PMID: 27534274).
4. Defective ciliary gate → **compromised docking/entry of intraflagellar transport (IFT) particles and impaired gating of membrane and soluble proteins → failure of axoneme assembly / primary-cilium biogenesis** (demonstrated, *C. elegans*).
5. In parallel (mammalian branch): HYLS1 is recruited to the ciliary base by **PIPKIγ** and **activates PIPKIγ by relieving its autoinhibitory dimerization**, expediting depletion of centrosomal PI(4)P to **permit axoneme nucleation** (demonstrated, mammalian cells; PMID: 34162535).
6. HYLS1 deficiency → **disrupted assembly of the ciliary NPHP module and agonist-induced ciliary exit of β-arrestin → impaired removal of ciliary GPR161 → dysregulated Hedgehog (Hh) signaling** (demonstrated, mammalian cells).
7. Defective primary-cilium–dependent signaling (Hedgehog central; likely Wnt/PCP contributions inferred) during early embryogenesis → **disrupted midline patterning, neural tube/forebrain development, and limb anterior-posterior patterning → clinical malformations**: hydrocephaly/hydranencephaly, absent midline brain structures, occipitoschisis, micrognathia, cleft lip/palate, polydactyly, and lung-lobation defects (inferred from cilium/Hedgehog biology linking to the observed phenotype).



Detail by category

- **Molecular pathways.** Primary-cilium–dependent **Hedgehog signaling** is the best-documented affected pathway (via GPR161/β-arrestin/NPHP module; PMID: 34162535). **Phosphoinositide metabolism (PI(4)P via PIPKIγ)** is the proximal lipid-signaling mechanism. Wnt/planar-cell-polarity involvement is plausible for ciliopathies but not directly demonstrated here (inferred).
- **Cellular processes.** Centriole/basal-body biogenesis, ciliogenesis, intraflagellar transport, and ciliary gating. GO suggestions: **GO:0060271 cilium assembly; GO:0098534 centriole assembly; GO:0035869 ciliary transition zone; GO:0036064 ciliary basal body; GO:0042073 intraflagellar transport; GO:0007224 smoothened signaling pathway**.
- **Protein dysfunction.** Loss of function via impaired protein–protein interaction (HYLS1–CEP120) and mislocalization; not aggregation. The centriole inner scaffold is destabilized.
- **Metabolic / biochemical.** Localized phosphoinositide (PI(4)P) remodeling at the centrosome mediated by lipid kinase PIPKIγ (CHEBI: phosphatidylinositol 4-phosphate). No systemic metabolic defect.
- **Immune involvement.** None.
- **Tissue damage mechanism.** Developmental (patterning failure) rather than degenerative/inflammatory injury.

- **Molecular profiling.** Honkala et al. performed molecular/cellular profiling of HLS fetuses, revealing "multiple changes in molecular and cellular level," including abnormal radial glial cells (PMID: 19400947). No large transcriptomic/proteomic/metabolomic/single-cell datasets specific to HLS1 are established beyond fetal pathology.

Cell types involved (CL suggestions): radial glial cell (CL:0000681), neural progenitor/neuroepithelial cell (CL:0000031), ciliated cell (CL:0000064), spermatozoon/male germ cell (CL:0000019) — HYLS1 is also critical for spermatogenesis/flagella in *Drosophila* (PMID: 32509774).

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7. Anatomical Structures Affected

- **Primary organs / systems:** Central nervous system (brain — forebrain midline, cerebral cortex, hypothalamus; UBERON:0000955 brain, UBERON:0001890 forebrain), cranium/skull base (occiput), lungs (UBERON:0002048), limbs/autopod (hands and feet; UBERON:0002398, UBERON:0002387), craniofacial structures (mandible UBERON:0001684; palate UBERON:0001708).
- **Secondary involvement:** Genitourinary (abnormal genitalia; kidney size asymmetry reported in a case).
- **Tissue/cell level:** Neuroepithelium and radial glia (CNS), ciliated epithelia; connective/skeletal tissue (polydactyly, skull defects).
- **Subcellular level (GO Cellular Component): centriole (GO:0005814), centrosome (GO:0005813), ciliary basal body (GO:0036064), ciliary transition zone (GO:0035869), cilium (GO:0005929), axoneme (GO:0005930)**. HYLS1 protein localizes to the centriole/basal body; mutant protein mislocalizes to nuclear structures.
- **Localization / lateralization:** Malformations are predominantly **midline and bilateral** (midline brain defects, cleft lip/palate, occipitoschisis), with bilateral polydactyly.

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8. Temporal Development

- **Onset: Congenital / prenatal** — malformations arise during embryogenesis and are detectable by first- and second-trimester ultrasound (as early as the 11th week in at-risk families; PMID: 10210131).

- **Onset pattern:** Non-applicable in the acute/chronic sense; structural anomalies are present from organogenesis.
- **Progression / course:** The disorder does not "progress" post-natally because it is **perinatally lethal** — stillbirth or death within hours to days of birth. A rare "milder" case survived to seven months (PMID: 10210131), attributed to allelic variability.
- **Duration:** Self-limited by lethality; not chronic.
- **Remission:** None. **Critical period:** the embryonic window of cilium-dependent midline/neural/limb patterning — the only "intervention window" is reproductive (preimplantation/prenatal).

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9. Inheritance and Population

- **Inheritance pattern:** Autosomal recessive.
- **Penetrance:** Complete (biallelic pathogenic genotype → disease).
- **Expressivity:** Largely consistent in Finnish D211G homozygotes; allelic variability can yield "milder"/atypical presentations (PMID: 10210131).
- **Founder effect:** Strong — the Finnish D211G allele is an archetype of a Finnish-disease-heritage founder mutation (~27-fold enrichment in Finns; Finding F006), identified by Finnish linkage/LD mapping (PMID: 15843405).
- **Consanguinity:** Increases recurrence risk for rare alleles outside Finland (standard for AR disorders).
- **Carrier / incidence estimates (derived, Finding F006):** Under Hardy–Weinberg, Finnish carrier frequency ≈ **1.70% (~1 in 59)**; predicted homozygous-affected incidence ≈ **7.3×10^{-5} (~1 in 13,600)** — the same order as the classically reported Finnish HLS incidence of ~1 in 20,000 births.
- **Geographic distribution:** Enriched in Finland; rare, sporadic cases reported elsewhere (France, Romania, Norway, non-Finnish families). D211G is concentrated in Finland; novel alleles reported in non-Finnish families.
- **Sex ratio:** No strong sex bias reported; abnormal genitalia noted in some fetuses.
- **Genetic anticipation / germline mosaicism:** Not applicable / not reported.

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10. Diagnostics

- **Prenatal imaging:** Fetal ultrasound is the frontline tool — detects hydrocephaly/hydranencephaly, absent midline structures, polydactyly, and skull defects; diagnosis possible in the first trimester (≈11 weeks) in families with a prior affected fetus (PMID: 10210131).
- **Fetal autopsy / pathology:** Confirms the pathognomonic occipitoschisis ("keyhole") skull-base defect, CNS midline defects, immature cortex, abnormal radial glia, hypothalamic hamartoma, defective lung lobation, and polydactyly (PMID: 19400947, PMID: 22029171, PMID: 39857086).
- **Genetic testing (confirmatory):** Targeted testing for the Finnish D211G allele in Finnish/at-risk families; **single-gene HYLS1 sequencing**, **gene panels** (lethal fetal malformation / ciliopathy panels that include HYLS1), and **WES/WGS** for non-Finnish or atypical cases. Chromosomal microarray detects the 11q deletion. Maternal serum AFP may be elevated with open defects.
- **Biomarkers / lab tests:** No specific circulating biomarker; diagnosis is imaging + molecular.
- **Clinical criteria:** Recognized by the characteristic malformation triad (hydrocephaly/absent midline brain + polydactyly + the keyhole occipital defect) plus molecular confirmation.
- **Differential diagnosis:** Meckel(-Gruber) syndrome (occipital encephalocele, cystic kidneys, polydactyly; PMID: 17389183), acrocallosal syndrome (possible allelic/overlapping midline + polydactyly phenotype; PMID: 10756349), trisomy 13, Smith-Lemli-Opitz syndrome, Ellis-van Creveld syndrome (PMID: 23276573), Joubert syndrome (HYLS1 is also a rare Joubert gene; PMID: 36580738), Bardet-Biedl, and oral-facial-digital syndrome. The keyhole/occipitoschisis defect is the most discriminating feature favoring HLS.
- **Screening:** Carrier screening (especially in Finnish populations) and cascade testing of relatives; preimplantation and prenatal testing for at-risk couples.

11. Outcome / Prognosis

The prognosis is **uniformly fatal**. Affected fetuses are stillborn or die within hours to days after birth; a single atypical "milder" case survived to seven months (PMID: 10210131). There is no survival benefit from any intervention. Mortality is 100% and disease-specific. Morbidity/

quality-of-life and disability measures are not applicable to surviving patients; the principal measurable burden is reproductive loss and family bereavement. The mouse disease-mutation model dies "shortly after birth," mirroring the human lethality (PMID: 40009365). **Prognostic factors:** genotype/allelic severity appears to modulate the rare atypical/"milder" survivors. No prognostic biomarkers exist.

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12. Treatment

There is no disease-modifying or curative treatment for HLS1. Because the disorder is a prenatal, uniformly lethal malformation syndrome, care is limited to:

- **Supportive / palliative perinatal care** (NCIT: Palliative Care; comfort care for the neonate and family).
- **Genetic counseling** for recurrence risk and reproductive planning.
- **Pregnancy management**, including the option of termination following prenatal diagnosis per local practice.

No pharmacotherapy, gene therapy, cell therapy, RNA-based therapy, targeted therapy, immunotherapy, surgery, or rehabilitation alters the outcome. No clinical trials target HLS1. Pharmacogenomics and combination/personalized-medicine strategies are not applicable. *Not available for this disease.*

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13. Prevention

Prevention is entirely **reproductive/genetic**, not clinical:

- **Primary prevention:** None for an established pregnancy. Pre-conception **carrier screening** (particularly for the Finnish D211G allele) identifies at-risk couples.
- **Secondary prevention / early detection: Prenatal ultrasound** (first-trimester feasible in at-risk families; PMID: 10210131) and **prenatal molecular testing** (CVS/amniocentesis).
- **Preimplantation genetic testing (PGT-M):** Available for couples who are known carriers.
- **Genetic counseling:** Central — 25% recurrence risk per pregnancy for carrier couples; cascade testing of relatives.

- **Immunization / behavioral / public-health / environmental interventions:** Not applicable.

14. Other Species / Natural Disease

HYLS1 is **evolutionarily conserved** across metazoans. Confirmed orthologs (Finding F008):

Species	Gene	NCBI GeneID
Human	HYLS1	219844
Mouse (*Mus musculus*)	Hyls1	76832
Rat (*Rattus norvegicus*)	Hyls1	680262
Caenorhabditis elegans	hyls-1	functional ortholog
Drosophila melanogaster	HYLS1 ortholog	functional ortholog

No naturally occurring HLS-equivalent disease is catalogued in companion animals or wildlife (no OMIA entry noted), and the condition has no zoonotic or cross-species transmission relevance. Comparative biology is driven by experimental models rather than spontaneous animal disease. The deep conservation of HYLS1's centriole/cilia function (worm → fly → mouse → human) underpins the translatability of model findings (PMID: 19656802, PMID: 27534274, PMID: 32509774).

15. Model Organisms

HLS1 is exceptionally well served by cross-species models, which collectively built the mechanistic chain:

Model	Type	Key contribution	PMID
Mouse (HYLS1 disease-mutation knock-in)	Mammalian, genetic	Dies shortly after birth; recapitulates HLS defects; established CEP120 recruitment and centriole-integrity mechanism	38496445, 40009365
C. elegans (hyls-1)	Invertebrate, genetic	Defined HYLS-1 role in centriole structure, transition fibre/zone, ciliary gating, IFT docking	19656802, 27534274
Drosophila	Invertebrate, genetic	Confirmed mutation significance; ciliogenesis and spermatogenesis/flagella roles	15843405, 32509774
Mammalian cell culture	In vitro	PIPKIγ activation, PI(4)P depletion, NPHP/β-arrestin/GPR161–Hedgehog axis	34162535

Phenotype recapitulation: The mouse disease-mutation model reproduces several human HLS manifestations and dies perinatally, making it the best in vivo surrogate; the worm/fly models recapitulate the cellular ciliogenesis defect but not the organismal malformation pattern. **Limitations:** Invertebrate models cannot reproduce mammalian forebrain/midline morphogenesis; even the mouse may not capture the full human craniofacial/skull-base phenotype (tissue-specific cilia dependence). **Resources:** MGI (mouse Hyls1 76832), RGD (rat Hyls1 680262), WormBase (hyls-1), FlyBase.

Mechanistic Model / Interpretation

The HLS1 literature now supports a **single, coherent, upstream-to-downstream mechanism** in which a point mutation in a centriole protein cascades into a signaling ciliopathy:

Centriole integrity (CEP120–HYLS1) → ciliary gate (TF/TZ) → primary cilium → Hedgehog signaling → embryonic patterning.

The two independent molecular branches discovered — (a) the **mouse CEP120/centriole-inner-scaffold** axis (PMID: 38496445) and (b) the **mammalian PIPKIγ/PI(4)P/GPR161–Hedgehog** axis (PMID: 34162535) — are complementary rather than contradictory: the first explains **why the cilium fails to assemble** (structural), the second explains **why cilium-dependent signaling fails even where cilia form** (functional/signaling). Both terminate in

defective Hedgehog-dependent midline, neural, and limb patterning, which is a classic explanation for the HLS malformation triad (absent midline brain structures, polydactyly, craniofacial clefts). The tissue-specificity of centriole/cilia dependence plausibly accounts for the particular organs affected.

Evidence Base

PMID	Title (abbrev.)	Role in this report
15843405	*HLS caused by a missense mutation in a novel gene HYLS1*	Gene discovery; D211G; mutant mislocalization; *Drosophila* confirmation
19656802	*HYLS-1 links core centriole structure to cilia formation*	Establishes ciliopathy; D211G impairs ciliogenesis; centriole anchoring
27534274	*HYLS-1 regulates formation of the ciliary gate*	TF/TZ formation, ciliary gating, IFT docking
19400947	*Unraveling the disease pathogenesis behind lethal HLS*	Core phenotype; CNS neuropathology; molecular/cellular profiling
22029171	*Lethal evolution of a newborn with HLS — Romanian patient*	Occipitoschisis as pathognomonic; craniofacial/limb features
34212369	*First two non-Finnish HYLS1 variants*	Novel alleles (R221P, R205*); phenotype expansion
38496445 / 40009365	*Centriole structural integrity defects in HLS*	Mouse model; CEP120 recruitment; centriole-integrity mechanism
34162535	*HYLS1 activates the ciliary lipid kinase PIPKIγ*	Lipid-kinase mechanism; GPR161/β-arrestin/Hedgehog
32509774	*Functional analysis of HYLS1 in ciliogenesis and spermatogenesis*	Conserved function; germ-cell/ flagella role
10210131	*HLS in a non-Finnish family; early prenatal diagnosis*	Allelic variability; first-trimester ultrasound
39857086	*Hydrolethalus Syndrome: a rare congenital disorder*	Keyhole defect, lung lobation, open-book hemispheres
17389183, 10756349, 23276573, 36580738	Differential diagnoses	Meckel, acrocallosal, Ellis-van Creveld, Joubert overlap

Evidence source mix: human clinical/fetal-pathology (phenotype, diagnosis, epidemiology), model organism (mechanism: mouse, worm, fly), in vitro (PIPKIγ/Hedgehog), and computational (gnomAD allele-frequency/Hardy-Weinberg epidemiology).

Limitations and Knowledge Gaps

- **Allelic spectrum is narrow.** The great majority of molecular evidence rests on the single Finnish D211G allele; generalizability to rare non-Finnish alleles (R221P, R205*, CNVs) is only partially characterized.
- **Phenotype–genotype modifiers unknown.** The basis for rare "milder"/atypical survivors and for the HYLS1–Joubert phenotype is not resolved; no validated modifier genes or epigenetic mechanisms exist.
- **No human omics at scale.** There are no large transcriptomic, proteomic, or single-cell datasets specific to HLS1 fetuses; mechanism relies on model systems.
- **Branch integration incomplete.** The precise quantitative contribution of the centriole-integrity branch vs the PIPKIγ/Hedgehog branch to each specific malformation (brain vs limb vs lung) is not experimentally dissected.
- **No therapeutic avenue.** Lethality and prenatal onset mean no treatment pipeline exists.

Proposed Follow-up Experiments / Actions

1. **Genotype–phenotype catalog:** Compile all reported HYLS1 alleles with precise fetal phenotypes to test whether null vs hypomorphic alleles predict HLS vs Joubert-spectrum outcomes.
2. **Conditional/tissue-specific mouse models:** Use the existing disease-mutation mouse to dissect tissue-specific cilia dependence (forebrain vs limb bud vs lung), clarifying which malformations are Hedgehog-driven.
3. **Single-cell / spatial transcriptomics of model embryos:** Map Hedgehog/GPR161 pathway dysregulation across developing midline, neural tube, and limb bud to connect molecular branches to specific defects.
4. **Structural biology of the HYLS1–CEP120 interface:** Determine how D211G, R221P, and R205* differentially disrupt CEP120 binding and centriole localization (AlphaFold-guided, then validated).

5. **Carrier-screening and epidemiology refinement:** Validate the Hardy-Weinberg-derived Finnish incidence against registry data and extend gnomAD-based carrier estimates to non-Finnish populations for counseling.
6. **Prenatal diagnostic standardization:** Formalize first-trimester ultrasound + targeted HYLS1 panels (including the 11q deletion) as a diagnostic algorithm for suspected lethal midline ciliopathy.

Report compiled from 8 confirmed findings and 21 reviewed papers over 5 investigation iterations. All mechanistic and clinical claims are attributed to primary literature by PMID; model-organism, in vitro, computational, and human-clinical evidence sources are distinguished throughout.
